

Spatial scRNA-seq

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Overview

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1. Abstract

2. Introduction

3. Materials

4. Methods

Methods I

Basic method

- 1 Scanpy (Wolf, Angerer, & Theis, 2018)

Projection method

- 1 UMAP (McInnes, Healy, & Melville, 2018; Becht et al., 2019)
- 2 tSNE (Maaten & Hinton, 2008)
- 3 PHATE (Moon et al., 2019)
- 4 trimap (Amid & Warmuth, 2019)

Clustering

- 1 Leiden (Traag, Waltman, & van Eck, 2019)
- 2 Louvain (Blondel, Guillaume, Lambiotte, & Lefebvre, 2008)

5. Results

5. Results

5.1. Quality check

Total counts per cell I

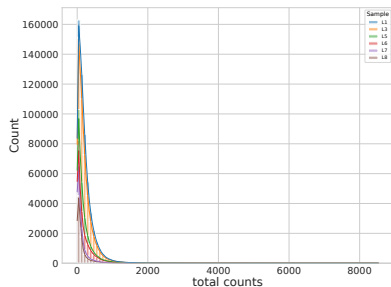


Figure: Total counts

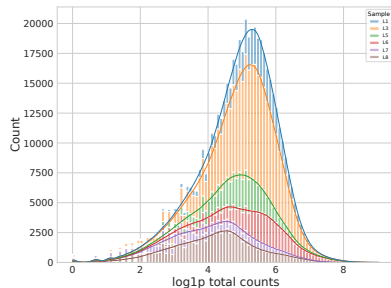


Figure: $\log_{1p}(\text{Total count})$

Total counts per cell II

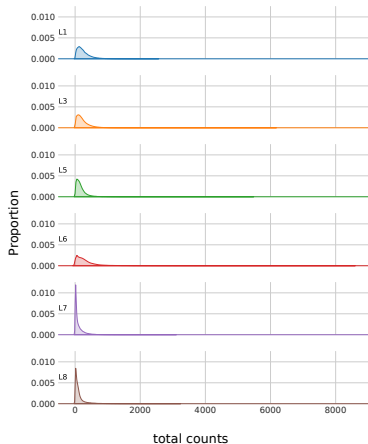


Figure: Genes by counts

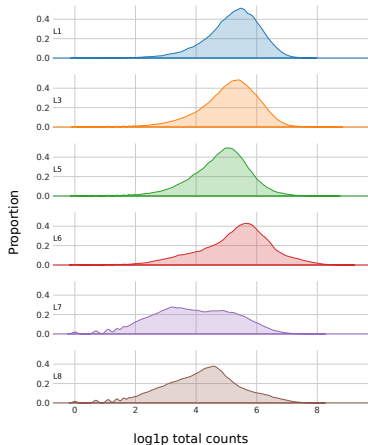


Figure: $\log_{1p}(\text{Genes by count})$

Total counts per cell III

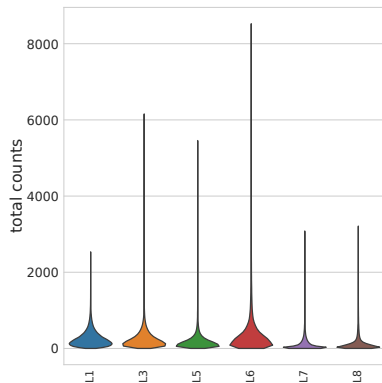


Figure: Genes by counts

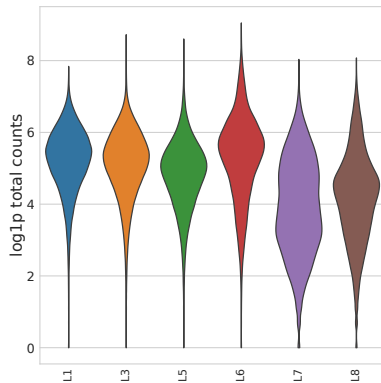


Figure: log1p(Genes by count)

Genes by counts I

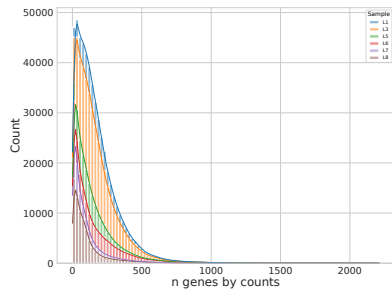


Figure: Genes by counts

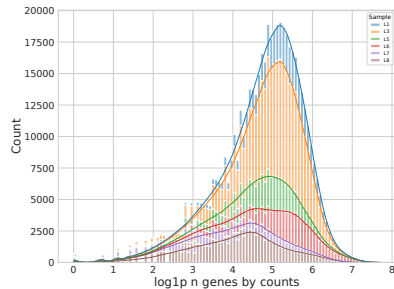


Figure: $\log_{1p}(\text{Genes by count})$

Genes by counts II

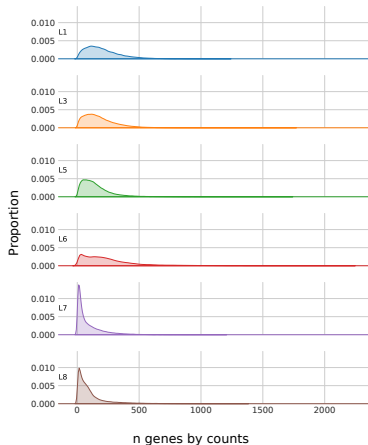


Figure: Genes by counts

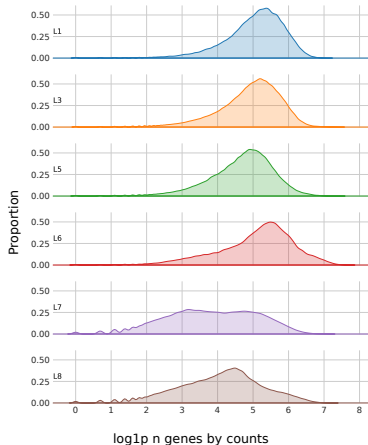


Figure: $\log_{1p}(\text{Genes by count})$

Genes by counts III

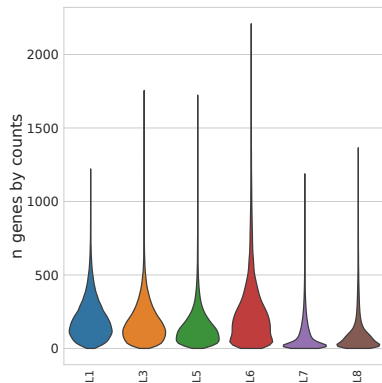


Figure: Genes by counts

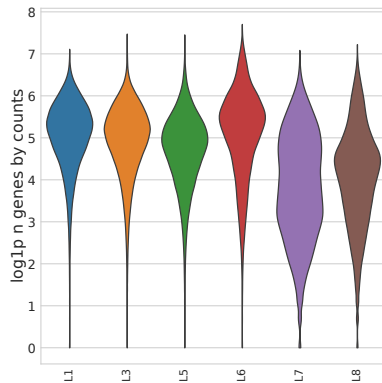


Figure: log1p(Genes by count)

5. Results

5.2. Highly variable genes

Highly variable genes I

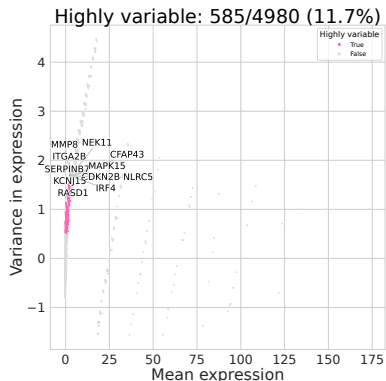


Figure: Volcano plot

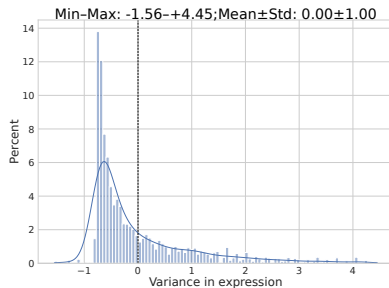


Figure: Histogram

Highly variable genes II

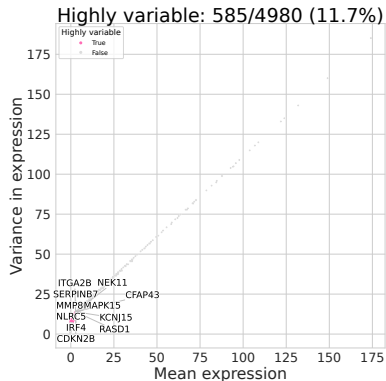


Figure: Volcano plot

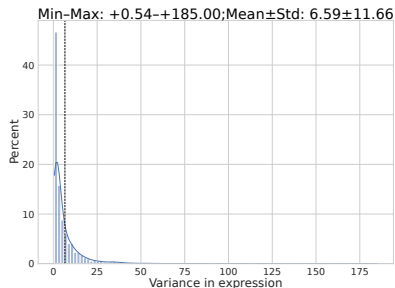


Figure: Histogram

5. Results

5.3. PCA

5. Results

5.4. Clustering

Clustering with UMAP I

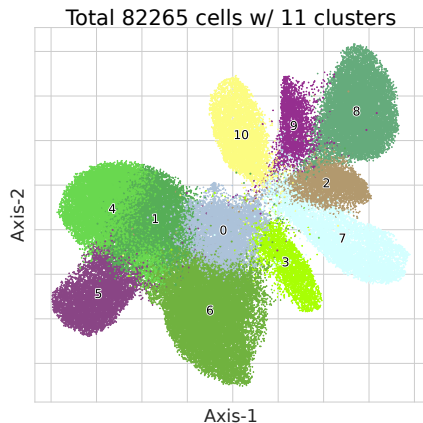


Figure: Clustering with UMAP

Clustering with UMAP II

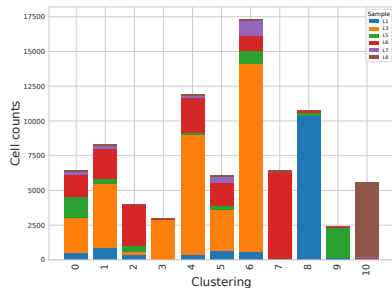


Figure: Absolute count

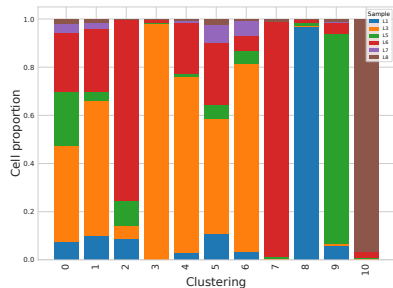


Figure: Relative proportion

Clustering with UMAP III

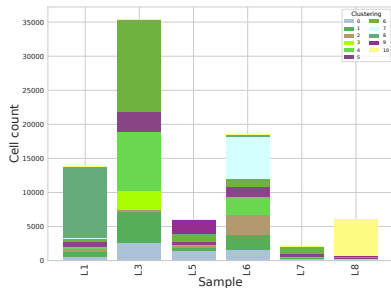


Figure: Absolute count

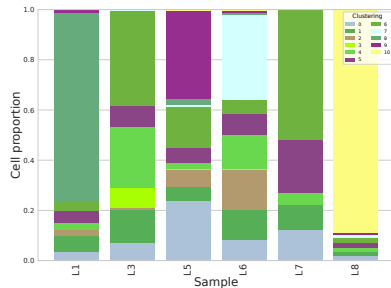


Figure: Relative proportion

5. Results

5.5. Marker genes

6. References

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